

# CS 486/686

# Introduction to Machine Learning

Yuntian Deng

---

Lecture 16

RN 19.1–19.2 · PM 7.1–7.2

Search ›

Uncertainty ›

Decisions ›

Learning

## Learning goals

- Identify reasons for building an agent that can **learn**.
- Describe different **types of learning** (supervised, unsupervised, RL).
- Define **supervised learning, classification, and regression**.
- Define **bias, variance**, and describe the trade-off.
- Use **cross-validation** to prevent overfitting.

# Why an agent that learns?



Medical  
diagnosis



Spam filtering



Face  
recognition



Speech



Handwriting

**Learning** = the ability to improve on future tasks based on experience.

- We can't anticipate every situation.
- We can't anticipate every change over time.
- For many tasks, we have no idea how to *program* a solution — only how to *show examples*.

# The learning architecture

Every learning system is defined by four ingredients:

## Task

The behaviour we want to improve (classify, predict, decide).

## Data

Experience used to improve (examples, demonstrations, rewards).

## Background knowledge / bias

What we assume up front (model class, smoothness, priors).

## Measure of improvement

How we score progress (accuracy, speed, new abilities).

# Three kinds of learning

## Supervised



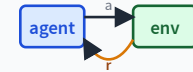
Given input + target features, predict targets for new inputs.  
Today's focus.

## Unsupervised



No targets — find structure: clustering, dimensionality reduction.  
L17.

## Reinforcement



Learn from rewards and punishments — what to *do*.  
L15.

# Supervised or unsupervised?

**Q1.** We have a user's credit-card transactions and want to flag any that look *different* from the rest. We have no fraud labels.

A. Supervised

B. **Unsupervised**

**B — Unsupervised.** No target labels to predict.

**Q2.** We have historical weather labels (sunny/cloudy/rain/snow) for a date, and we want to predict the same date next year.

A. **Supervised**

B. Unsupervised

**A — Supervised.** Each example has a target (the weather label).

# Two flavors of supervised learning

## Classification



Target features are **discrete** — e.g. dog vs cat.

## Regression



Target features are **continuous** — e.g. tomorrow's temperature.

# Quick check: classification or regression?

**Q3.** Historical weather labels (sunny / cloudy / rain / snow); predict next year's label for a given date.

A. **Classification**

B. Regression

**A — Classification.** The target takes a discrete value from a fixed set.

**Q4.** Historical house prices over time; predict next month's price for a particular house.

A. Classification

B. **Regression**

**B — Regression.** The target is a continuous real number.



# Supervised learning = function approximation

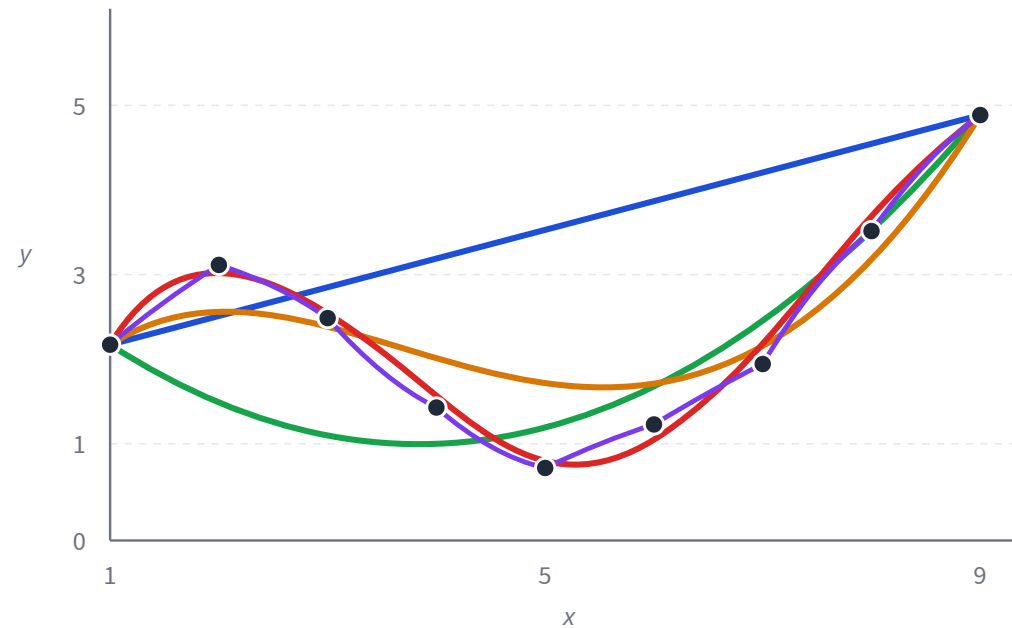
Given training examples  $(x_i, f(x_i))$ , find a **hypothesis**  $h$  that approximates the unknown true function  $f : X \rightarrow Y$ .

A **hypothesis space** is the set of functions we'll consider (e.g. all polynomials of degree  $\leq k$ ).

Learning = a search problem in the hypothesis space — usually some form of local search (gradient descent, etc.).

# Example: fitting a polynomial

Same 9 data points, five candidate polynomial models. Watch how the fit changes with complexity.



— degree 1 (line)    — degree 2    — degree 3    — degree 5    — degree 9 (overfits)

# Which curve is "right"?

- If the data has **outliers** or is noisy, simpler curves are more trustworthy.
- If the data really does have wild swings (stock prices, etc.), the complex curve is closer to truth.
- **There is no perfect answer** — each curve is justified *by some assumption*.

## No-free-lunch

To learn something useful, we *must* make assumptions. Those assumptions are the **inductive bias** of the learner.

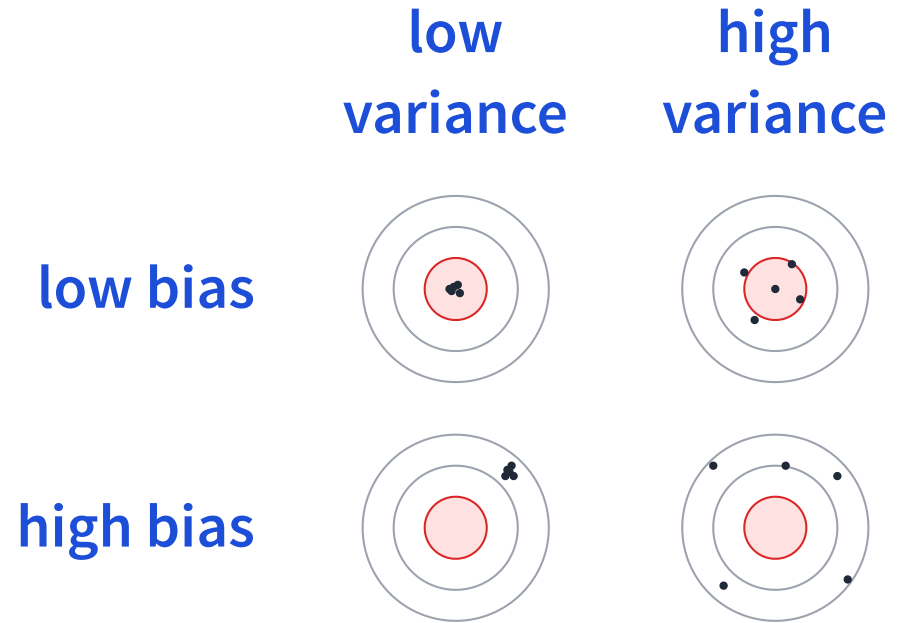
# Bias and variance

## Bias

With infinite data, how well can the family fit  $f$ ?  
High bias = too simple, **underfits**.

## Variance

How much does the learned hypothesis change with different training sets? High variance = too flexible, **overfits**.



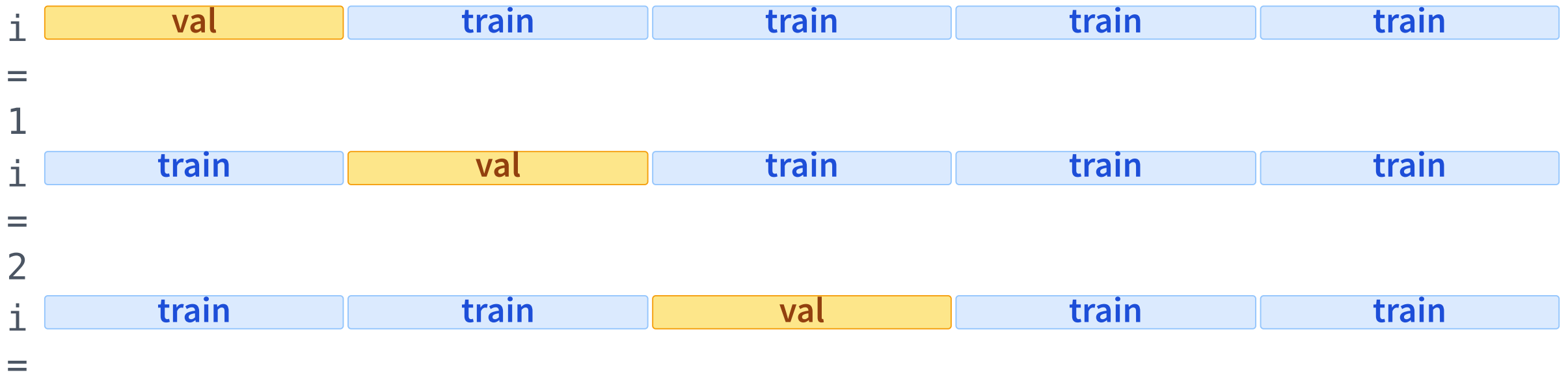
Dots = predictions from re-trained models.  
Bullseye = truth.

# K-fold cross-validation

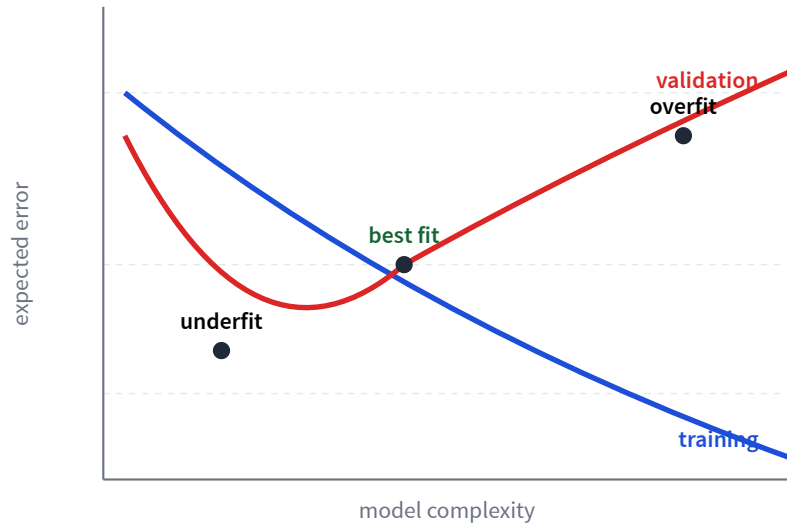
How to pick a hypothesis that generalizes? Use a slice of training data as a **surrogate test set**.

k-fold-CV(model, dataset,  $K$ )

1. Split training data into  $K$  equally sized folds.
2. For  $i = 1, \dots, K$ : train on the other  $K - 1$  folds; evaluate on fold  $i$  (held-out validation).
3. Average error over the  $K$  runs.



# Overfitting: training vs validation error



**Q5.** Which is more likely to overfit?

- A. Straight line (simple)
- B. 4th-degree polynomial (complex)
- C. I don't know

**B — complex.** More degrees of freedom = more wiggle = it can chase the noise.

# True error vs empirical error

Setup: domain  $X$ , label space  $Y$ , unknown true function  $f : X \rightarrow Y$ . Examples drawn from distribution  $D$  over  $X$ .

## True error (what we want)

$$L_{D,f}(h) = \Pr_{x \sim D} [h(x) \neq f(x)]$$

The probability of being wrong on a fresh random example. We **can't** compute this — we don't know  $D$  or  $f$ .

## Empirical error (what we measure)

$$L_S(h) = \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m}$$

# Empirical risk minimization (ERM)

Given a hypothesis class  $\mathcal{H}$  and a loss  $\ell$ , pick the hypothesis with smallest training loss:

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i)$$

Three factors determine how well ERM generalizes:

## Data size $m$

Larger is better: more examples  $\Rightarrow$  closer empirical  $\rightarrow$  true.

## Hypothesis space $\mathcal{H}$

Trade-off: small  $\mathcal{H}$  = high bias; large  $\mathcal{H}$  = high variance.

## Loss function $\ell$

Encodes what kind of error you care about (0/1, squared, hinge, ...).



## Learning goals (recap)

- ✓ Identify reasons to build a learning agent.
- ✓ Describe different **types of learning**.
- ✓ Define **supervised learning, classification, regression**.
- ✓ Define **bias, variance**, and their trade-off.
- ✓ Use **cross-validation** to prevent overfitting.

## Next: unsupervised learning

Today: labels guided every hypothesis we picked. Without labels, what *structure* can we still find in data?

L17: clustering, k-means, autoencoders.